COMP3330 A2: Machine Intelligence

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Introduction

The UCI Machine Learning Repository [1] maintains 678 datasets collated from differing sources, individually built using differing parameters and sometimes thousands of sample points. Within this collection is the Iris dataset constructed by Robert A. Fisher in 1936, which is one of the earliest known datasets used for classification of Iris flowers using Multivariate Analysis and only contains 150 samples [2]. This dataset allows exploration into machine learning methodology as an educational tool, since the small number of sample points will allow easier manual checking to ensure designed algorithms are performing correctly.

Methodology

Robert A. Fisher opened many new areas in Multivariate Analysis, one of which is his research on discriminate analysis [3]. Discriminant function analysis is the statistical analysis used to analyse data when the dependent variable or outcome is categorical and independent variable or predictor variable is parametric [4]. In short, if the dependent variable is a dichotomy, there is one discriminant function; if there are k levels of the dependent variable, up to k – 1 discriminant functions can be extracted, and the useful projections can be retained [5].

This approach to classification fits very well into the Decision Tree Classifier model, which analyses the entire dataset at the root node, and repeatedly splits the data into smaller subsets based on the most informative features, until a stopping criterion is met [6]. It can also model complex, non-linear relationships between predictor variables and class variables, and they are non-parametric, meaning data does not require standardised or normalised distributions [7].

Implementation

The code used for investigation has been developed in an intentionally naive way for use of reading by people unfamiliar with Machine Learning code, or code in general. This aids in understanding, appropriation, and troubleshooting for future improvement.

Project code and data can be found here: https://github.com/katemcalpine/Iris-Classification.git

Project references can be found on the project GitHub, however some of the information in the reference links are incorrect or outdated. Sourced fixes were found to address the theoretical and programming issues, bringing the finished project up to date with correct information and current components. Specifications used for the project are outlined in table 1.

Language	Python 3.13
Platform	PyCharm Community Edition 2024.3.1.1
Dataset	Iris
Model	Decision Tree Classifier
Libraries	• pandas 2.2.3 • numpy 2.2.6 • matplotlib 3.10.3 • seaborn 0.13.2 • os • graphviz 0.20.3 • scikit-learn 1.6.1
Code	Various references found in the project GitHub

Table 1: Specifications Used for Project

Results and Discussion

Since the Iris dataset contains 150 samples, each with 5 recorded parameters, a correlation heat map will help gauge some understanding of how the dataset relates to each other, as shown in figure 1.

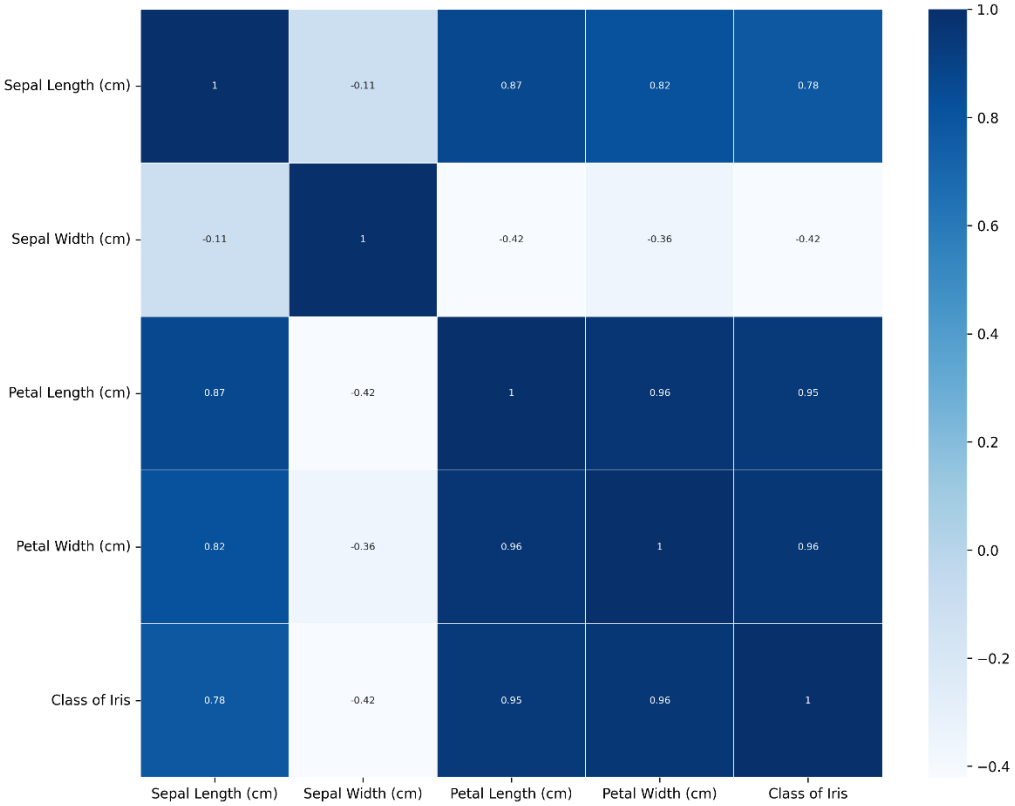


Figure 1: Iris Dataset Correlation Heat Map

When classifying data, preference is given to beneficial linear relationships. The parameters associated with highest importance sometimes have correlation coefficients with the highest magnitude or the greatest range. From the correlation heat map, we can see the parameters 'Sepal Width (cm)' vs. 'Class of Iris' and, 'Sepal Width (cm)' vs. 'Petal Length (cm)' share the same negative correlation coefficient of -0.42. 'Petal Length (cm)' also contains the largest range of coefficients (from -0.42 to 0.95), so may have the most complex relationships with the other parameters.

Upon further inspection of these identified parameters, a clarification starts to form as shown in figures 2 and 3.

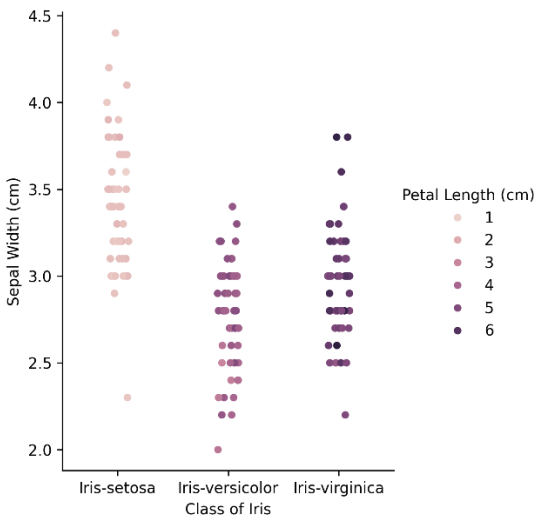
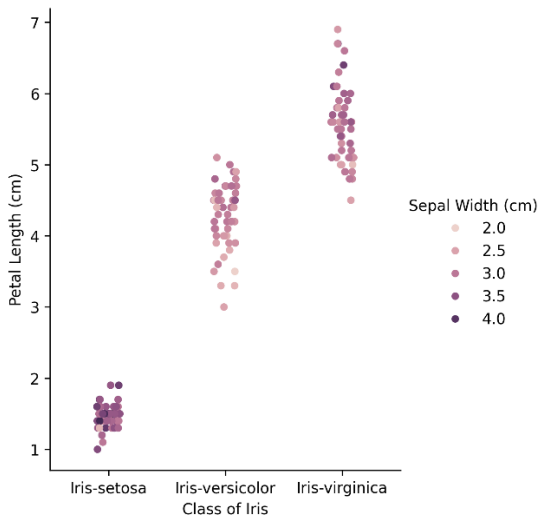


Figure 2a: 'Sepal Width (cm)' vs. 'Class of Iris'



b: ‘Petal Length (cm) vs. ‘Class of Iris’

In figure 2a, the class groupings appear to be separate, but the petal length colourations appear strongly defined between class grouping, reflecting the stronger linear relationship between class and petal length at a coefficient of 0.95. Figure 2b investigates these relationships further, noting the convolution of colours between classes as plotted by the sepal width. Yet again, the graph has favoured the strong relationship between class and petal length.

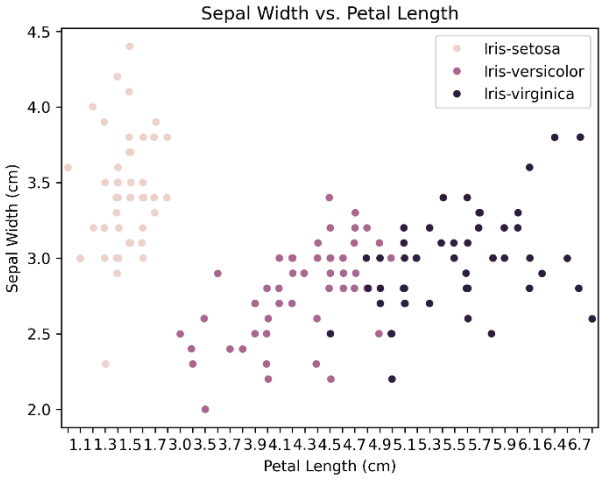


Figure 3: 'Sepal Width (cm)' vs. 'Petal Length (cm)'

Sepal width vs. petal lengths and sepal width vs. class have the same negative coefficients, so the definition is in the positive coefficient between petal length and class. This is reinforced in figure 3, which shows a clear overlap in class classification between petal length 4.5cm to 5.1cm and sepal width 2.5cm to 3.1cm, thus isolating and defining a strong relationship to use in classification. Through this investigation, it seems the parameter 'Petal Length (cm)' has the most influence and therefore may be the most important for classification.

Since the project goal is predicting the class of Iris flower, the focus was placed on the "Class of Iris" data column to set the x and y-axis and split data into seeded (to help with reproducibility) 80% train and 20% test subsets respectively to use in the Decision Tree Classifier. The result was then used to test the hypothesised parameter importance, confirming 'Petal Length (cm)' being the main influence for decision-making in the classification model.

Figure 4 shows the result of the Iris dataset with a training subset of 120 samples, and a test subset of 30 samples. The Gini coefficient reaches zero by the terminal leaf nodes, indicating full removal of disorders from the nodes and all instances belonging to the same class, thus classifying the data.

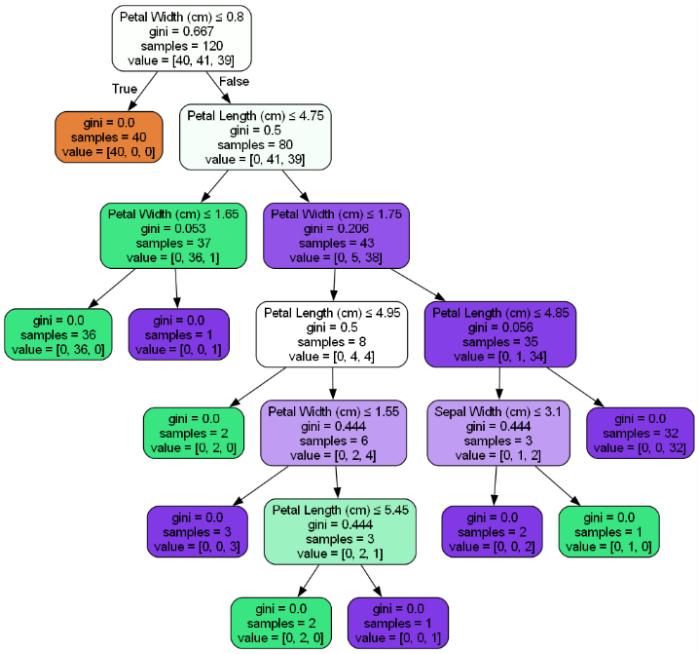


Figure 4: Decision Tree Classifier Result on Iris Dataset

Adding up the data from the coloured terminal leaf nodes, all training samples are accounted for but there is a clear discrepancy, as outlined in table 2.

Colour of Terminal Leaf Node	Resulting Sample Size
Orange	40
Green	41
Purple	39

Table 2: Analysing Data Groups from Decision Tree Classifier

The data in table 2 reveals an uneven number of training samples, which should not occur with an accurate model since the Iris dataset very clearly contains an even distribution of samples across all three Iris flower classes.

Test accuracy results in 100% when calculated in Python, but the Confusion Matrix in figure 6 suggests an Iris-versicolor may have been inaccurately classed as an Iris-virginica, thus only correctly classifying one of three classes of Iris. Pairing table 2 and the test samples from the Confusion Matrix, we can speculate that the orange node in the Decision Tree Classification is Iris-setosa, the green node is Iris-virginica, and the purple node is Iris-versicolor.

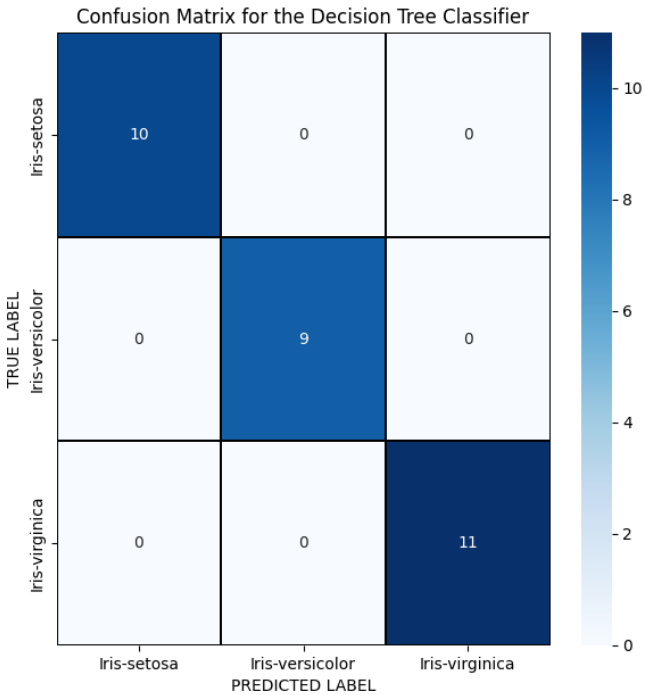


Figure 5: Confusion Matrix for the Decision Tree Classifier

Confirming the test accuracy results:

$$accuracy = \left(\frac{10 + 9 + 11}{30}\right) 100 = 100$$

This shows a false positive in the accuracy of this Decision Tree Classifier model. When testing an increased training subset size, the Confusion Matrix resulted in more errors, but the accuracy still calculated to 100%.

Conclusion and Future Work

The Decision Tree Classifier can be used to make decisions in many areas including operations, budget planning, and project management, but is a rather aggressive machine learning algorithm in its decision-making. This makes way for overfitting through noise integration and irrelevant patterns. While something like a neural network could increase the accuracy reflected in the confusion matrix, the ideal model would be multi-agent machine learning and acting through a complex network. This would help average out the assumptions and inaccuracies generated within limited models, hopefully toward a positive trajectory of clarification and confidence.

References

[1] UC Irvine, “UC Irvine Machine Learning Repository,” [Online]. Available: https://archive.ics.uci.edu/. [Accessed May 2025].

[2] R. A. Fisher, “Iris - UCI Machine Learning Repository,” [Online]. Available: https://archive.ics.uci.edu/dataset/53/iris. [Accessed May 2025].

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[4] M. K. G. R. P. J. S. S. K. B. Deepak Dhamnetiya, “How to Perform Discriminant Analysis in Medical Research? Explained with Illustrations,” 2022.

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